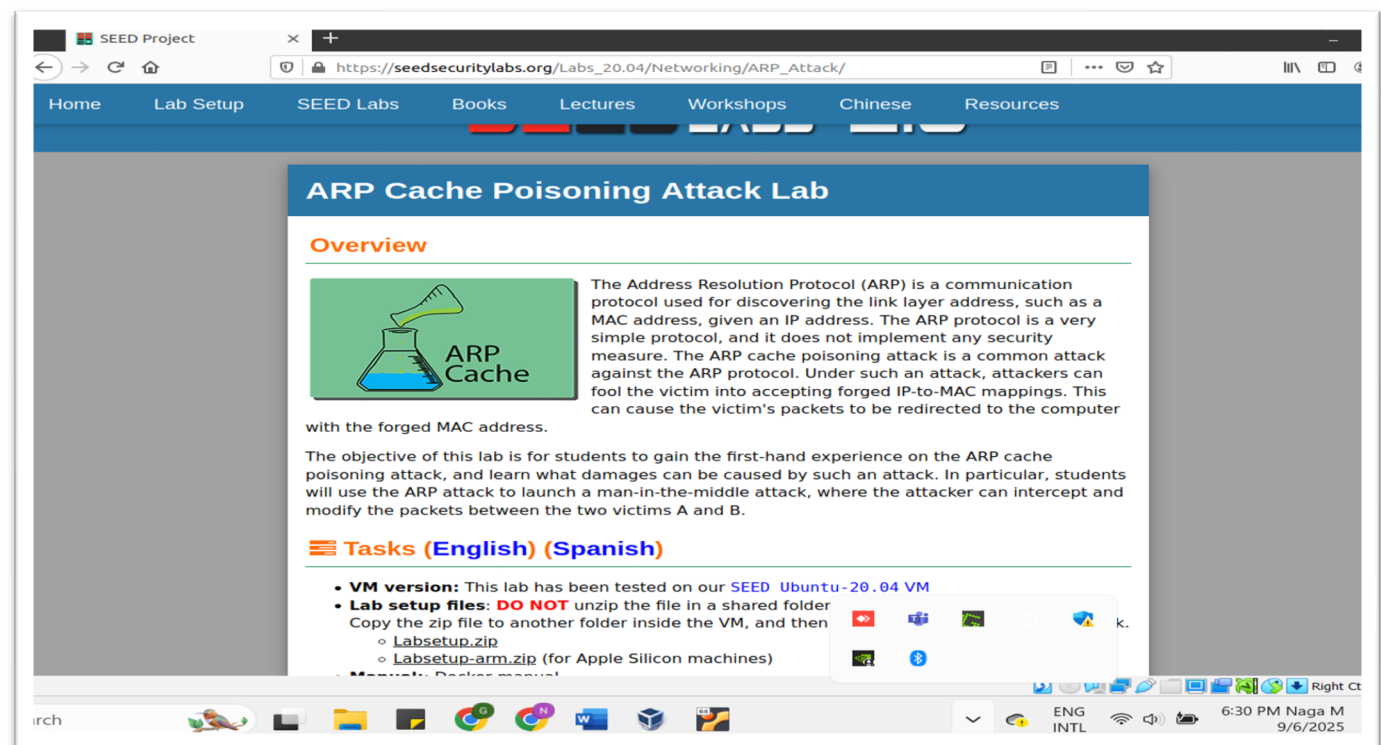


ARP ATTACK

Download the lab setup zip file.>> click on the seed labs >> network security>> ARP cache poisoning attack



ARP cache poisoning attack



Copy this zip to documents >> arpattack>> there unzip the labsetup.zip file. From here right click and open the Terminal then we can directly move to Terminal

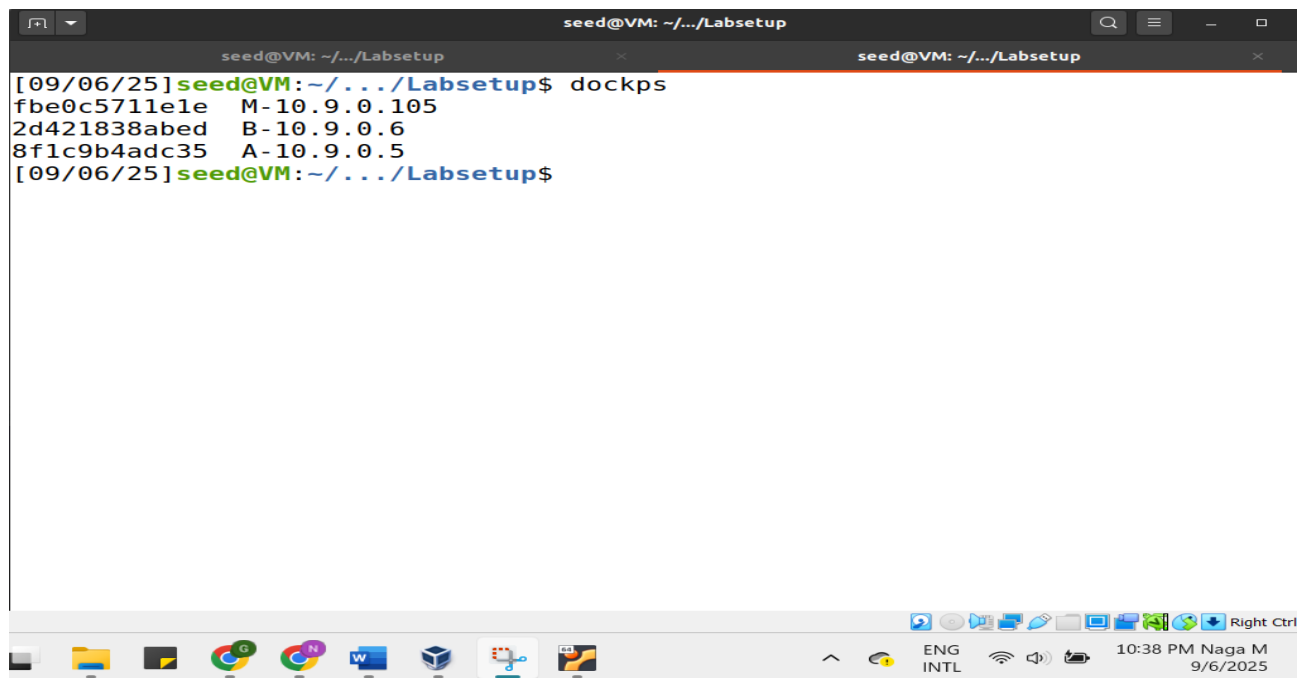
```
seed@VM: ~/.../Labsetup
[09/06/25]seed@VM:~/.../ARPatAttack$ cd labsetup
bash: cd: labsetup: No such file or directory
[09/06/25]seed@VM:~/.../ARPatAttack$ cd Labsetup
[09/06/25]seed@VM:~/.../Labsetup$
```

Now, We need to Set up Containers, by using the commands : dcbuild, dcup

```
seed@VM: ~/.../Labsetup
[09/06/25]seed@VM:~/.../ARPatAttack$ cd labsetup
bash: cd: labsetup: No such file or directory
[09/06/25]seed@VM:~/.../ARPatAttack$ cd Labsetup
[09/06/25]seed@VM:~/.../Labsetup$ dcbuild
HostA uses an image, skipping
HostB uses an image, skipping
HostM uses an image, skipping
[09/06/25]seed@VM:~/.../Labsetup$ dcup
Creating network "net-10.9.0.0" with the default driver
Creating B-10.9.0.6 ... done
Creating M-10.9.0.105 ... done
Creating A-10.9.0.5 ... done
Attaching to M-10.9.0.105, A-10.9.0.5, B-10.9.0.6
A-10.9.0.5 | * Starting internet superserver inetd [ OK ]
B-10.9.0.6 | * Starting internet superserver inetd [ OK ]
```

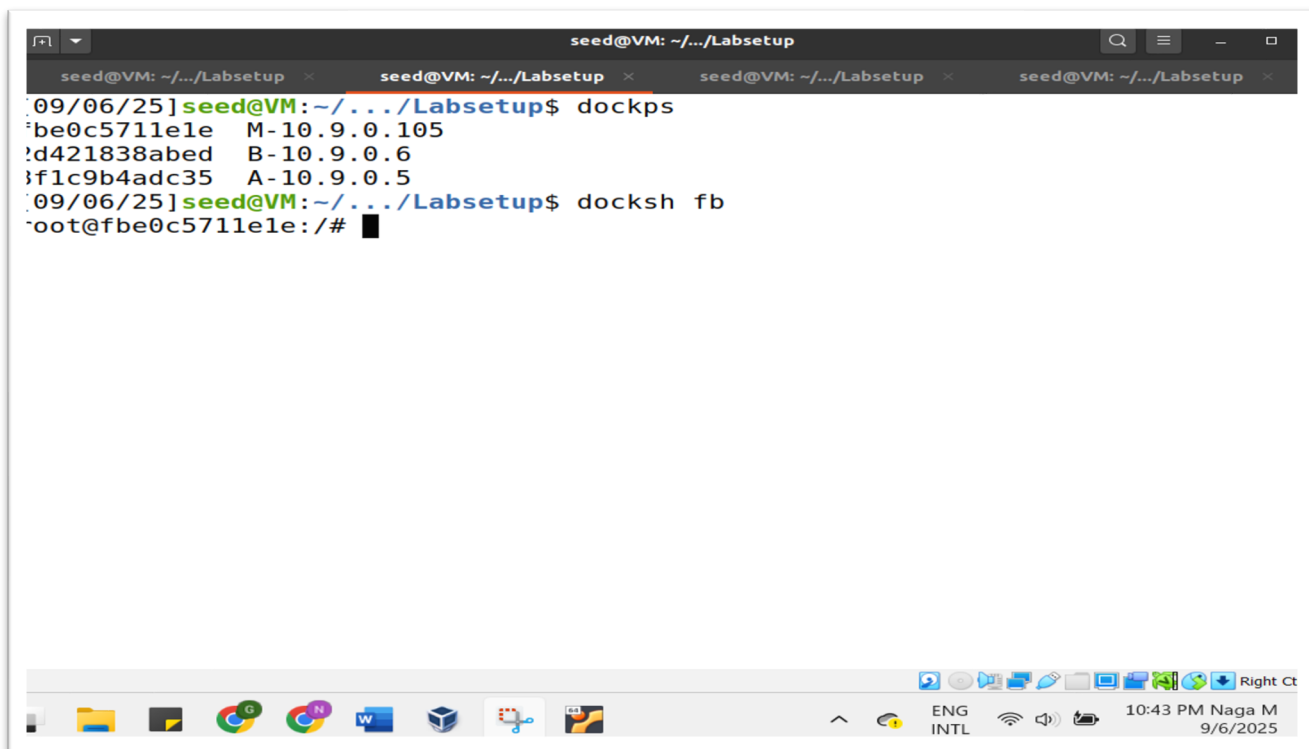
Here, from the output we can see the containers as like Host A: 10.9.0.5 , Host B: 10.9.0.6, Attacker-M: 10.9.0.105.

Open another terminal and in that type dockps



```
seed@VM: ~/.../Labsetup
[09/06/25] seed@VM: ~/.../Labsetup$ dockps
fbe0c5711e1e  M-10.9.0.105
2d421838abed  B-10.9.0.6
8f1c9b4adc35  A-10.9.0.5
[09/06/25] seed@VM: ~/.../Labsetup$
```

Here we can see that the attacker container and that id is fb, So we can run the command docksh fb in one terminal , as like open two more terminals and run docksh command for the Host A, Host B.



```
seed@VM: ~/.../Labsetup
[09/06/25] seed@VM: ~/.../Labsetup$ dockps
fbe0c5711e1e  M-10.9.0.105
2d421838abed  B-10.9.0.6
8f1c9b4adc35  A-10.9.0.5
[09/06/25] seed@VM: ~/.../Labsetup$ docksh fb
root@fbe0c5711e1e: /#
```

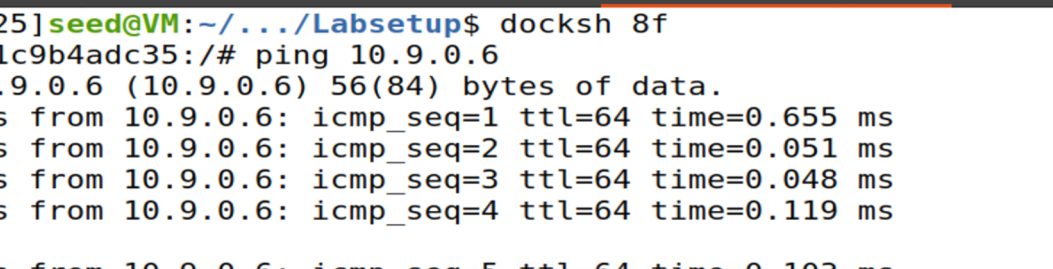
Lets See the below, left side is Host A, right side is Host B



The screenshot shows a terminal window with the title bar "seed@VM: ~/.../Labsetup". The terminal content displays the command "99/06/25]seed@VM:~/.../Labsetup\$ docksh 8f" and the output "ot@8f1c9b4adc35:/#". The terminal window is part of a desktop environment with a taskbar at the bottom showing various application icons and system status information including "ENG INTL", "10:45 PM", and "Naga M 9/6/2025".

The screenshot shows a terminal window with a dark theme. The title bar at the top reads "seed@VM: ~/.../Labsetup". The terminal content shows a prompt "09/06/25]seed@VM:~/.../Labsetup\$" followed by the command "docker sh". The prompt changes to "oot@2d241838abed:/#" after the command is executed. The bottom of the image shows a Windows taskbar with various application icons and a system tray on the right displaying the date and time as "10:47 PM Naga M 9/6/2025".

Lets check the LAN is working or not by ping Host B from Host A and vice versa.



The screenshot shows a terminal window with the title bar "seed@VM: ~/.../Labsetup". The terminal output is as follows:

```
09/06/25]seed@VM:~/.../Labsetup$ docksh 8f
oot@8f1c9b4adc35:/# ping 10.9.0.6
ING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.
4 bytes from 10.9.0.6: icmp_seq=1 ttl=64 time=0.655 ms
4 bytes from 10.9.0.6: icmp_seq=2 ttl=64 time=0.051 ms
4 bytes from 10.9.0.6: icmp_seq=3 ttl=64 time=0.048 ms
4 bytes from 10.9.0.6: icmp_seq=4 ttl=64 time=0.119 ms

4 bytes from 10.9.0.6: icmp_seq=5 ttl=64 time=0.103 ms
4 bytes from 10.9.0.6: icmp_seq=6 ttl=64 time=0.086 ms
4 bytes from 10.9.0.6: icmp_seq=7 ttl=64 time=0.098 ms
Z
1]+ Stopped                               ping 10.9.0.6
oot@8f1c9b4adc35:/#
```

The terminal window has four tabs, all titled "seed@VM: ~/.../Labsetup". The taskbar at the bottom shows various application icons including Google Chrome, Microsoft Word, and a file explorer, along with system status icons on the right.

```
seed@VM: ~/.../Labsetup
[09/06/25]seed@VM:~/.../Labsetup$ docksh 2d
root@2d421838abed:/# ping 10.9.0.5
PING 10.9.0.5 (10.9.0.5) 56(84) bytes of data.
64 bytes from 10.9.0.5: icmp_seq=1 ttl=64 time=0.228 ms
64 bytes from 10.9.0.5: icmp_seq=2 ttl=64 time=0.112 ms
64 bytes from 10.9.0.5: icmp_seq=3 ttl=64 time=0.098 ms
64 bytes from 10.9.0.5: icmp_seq=4 ttl=64 time=0.053 ms

64 bytes from 10.9.0.5: icmp_seq=5 ttl=64 time=0.105 ms
^Z
[1]+  Stopped                  ping 10.9.0.5
root@2d421838abed:/#
```

```
seed@VM: ~/.../Labsetup
[09/06/25]seed@VM:~/.../Labsetup$ docksh fb
fbee0c5711e1e M-10.9.0.105
2d421838abed B-10.9.0.6
3f1c9b4adc35 A-10.9.0.5
[09/06/25]seed@VM:~/.../Labsetup$ docksh fb
root@fbee0c5711e1e:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.9.0.105 netmask 255.255.255.0 broadcast 10.9.0.255
    ether 02:42:0a:09:00:69 txqueuelen 0 (Ethernet)
    RX packets 70 bytes 7348 (7.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@fbee0c5711e1e:/#
```

Here, we can confirm that it was eth0 and lets run the tcpdump on M-container... by using the tcpdump command as shown below.

```
seed@VM: ~/.../Labsetup
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@fbee0c5711e1e:/# tcpdump -i eth0 -n -e arp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
^Z
[1]+ Stopped tcpdump -i eth0 -n -e arp
root@fbee0c5711e1e:/# tcpdump -i eth0 -n
bash: tcpdump -i: command not found
root@fbee0c5711e1e:/# tcpdump -i eth0 -n
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
```

Above, we can see that it started listening, so immediately open Host A terminal and ping Host B..

```
seed@VM: ~/.../Labsetup
-6 use IPv6
-F <flowlabel> define flow label, default is random
-N <nodeinfo opt> use icmp6 node info query, try <help> as argument

For more details see ping(8).
root@8f1c9b4adc35:/# ping 10.9.0.6
PING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.
64 bytes from 10.9.0.6: icmp_seq=1 ttl=64 time=0.396 ms
64 bytes from 10.9.0.6: icmp_seq=2 ttl=64 time=0.128 ms
64 bytes from 10.9.0.6: icmp_seq=3 ttl=64 time=0.118 ms
64 bytes from 10.9.0.6: icmp_seq=4 ttl=64 time=0.092 ms
64 bytes from 10.9.0.6: icmp_seq=5 ttl=64 time=0.132 ms
64 bytes from 10.9.0.6: icmp_seq=6 ttl=64 time=0.149 ms
64 bytes from 10.9.0.6: icmp_seq=7 ttl=64 time=0.061 ms
64 bytes from 10.9.0.6: icmp_seq=8 ttl=64 time=0.114 ms
64 bytes from 10.9.0.6: icmp_seq=9 ttl=64 time=0.058 ms
64 bytes from 10.9.0.6: icmp_seq=10 ttl=64 time=0.113 ms
64 bytes from 10.9.0.6: icmp_seq=11 ttl=64 time=0.102 ms
64 bytes from 10.9.0.6: icmp_seq=12 ttl=64 time=0.094 ms
64 bytes from 10.9.0.6: icmp_seq=13 ttl=64 time=0.099 ms
64 bytes from 10.9.0.6: icmp_seq=14 ttl=64 time=0.126 ms
64 bytes from 10.9.0.6: icmp_seq=15 ttl=64 time=0.068 ms
64 bytes from 10.9.0.6: icmp_seq=16 ttl=64 time=0.081 ms
64 bytes from 10.9.0.6: icmp_seq=17 ttl=64 time=0.139 ms
```

Lets see the M-container Terminal now,

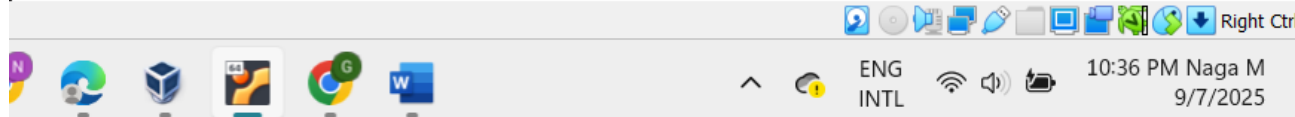
```
root@fbee0c5711e1e:/# tcpdump -i eth0 -n
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
04:34:03.561306 IP 10.9.0.5 > 10.9.0.6: ICMP echo request, id 31, seq 1, length 64
04:43:20.631304 IP fe80::42:2dff:fe1d:710b > ff02::2: ICMP6, router solicitation, length 16
^Z
[2]+ Stopped tcpdump -i eth0 -n
root@fbee0c5711e1e:/#
```

Task1: ARP Cache Poisoning

Lets see ,How to construct an ARP packet using Scapy. We can check with arp command from Host A, Host B

```
seed@VM: ~/.../Labsetup
[09/07/25]seed@VM:~/.../Labsetup$ docksh 31
root@31508b9ce735:/# arp -an
root@31508b9ce735:/# ping 10.9.0.6
PING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.
64 bytes from 10.9.0.6: icmp_seq=1 ttl=64 time=0.096 ms
64 bytes from 10.9.0.6: icmp_seq=2 ttl=64 time=0.065 ms
64 bytes from 10.9.0.6: icmp_seq=3 ttl=64 time=0.215 ms
64 bytes from 10.9.0.6: icmp_seq=4 ttl=64 time=0.101 ms
64 bytes from 10.9.0.6: icmp_seq=5 ttl=64 time=0.108 ms
^Z
[1]+  Stopped                  ping 10.9.0.6
root@31508b9ce735:/# arp -an
? (10.9.0.6) at 02:42:0a:09:00:06 [ether] on eth0
root@31508b9ce735:/#
```

```
seed@VM: ~/.../Labsetup
[09/07/25]seed@VM:~/.../Labsetup$ docksh 1c
root@1c7f4b0f814c:/# arp -an
root@1c7f4b0f814c:/# ping 10.9.0.5
PING 10.9.0.5 (10.9.0.5) 56(84) bytes of data.
64 bytes from 10.9.0.5: icmp_seq=1 ttl=64 time=0.056 ms
64 bytes from 10.9.0.5: icmp_seq=2 ttl=64 time=0.057 ms
64 bytes from 10.9.0.5: icmp_seq=3 ttl=64 time=0.124 ms
64 bytes from 10.9.0.5: icmp_seq=4 ttl=64 time=0.107 ms
64 bytes from 10.9.0.5: icmp_seq=5 ttl=64 time=0.056 ms
^Z
[1]+  Stopped                  ping 10.9.0.5
root@1c7f4b0f814c:/# arp -an
? (10.9.0.5) at 02:42:0a:09:00:05 [ether] on eth0
root@1c7f4b0f814c:/#
```



Task 1: ARP Cache Poisoning: In the attacker machine type scapy you can jum into scapy window, there we can see the class attributes of ether and Arp

```

seed@VM: ~/.../Labsetup
P////YCpc      A//A
scccccp///pSP///p    p//Y
sY//////////y  caa    S//P
cayCyayP//Ya      pY/Ya
sY/PsY////////YCc    aC//Yp
sc  sccaCY//PCypaapyCP//YsS
      spCPY////////YPSps
      ccaacs

Craft packets before they c
you.
-- Soc

>>> ls(ARP)
hwtype      : XShortField          = (1)
ptype       : XShortEnumField      = (2048)
hwlen       : FieldLenField        = (None)
plen        : FieldLenField        = (None)
op          : ShortEnumField        = (1)
hwsrc       : MultipleTypeField     = (None)
psrc        : MultipleTypeField     = (None)
hwdst       : MultipleTypeField     = (None)
pdst        : MultipleTypeField     = (None)
>>> ls(Ether)
dst         : DestMACField          = (None)
src         : SourceMACField        = (None)
type        : XShortEnumField      = (36864)

```

Lets find the Mac address for M-attacker by using the ipconfig command in the machine(attacker) terminal. From the output we can see that **M's Mac is: "02:42:0a:09:00:69"**

```

seed@VM: ~/.../Labsetup
root@fed47278651b:/# nano arp_poision.py

Use "fg" to return to nano.

[1]+  Stopped                  nano arp_poision.py
root@fed47278651b:/# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.9.0.105 netmask 255.255.255.0  broadcast 10.9.0.255
    ether 02:42:0a:09:00:69 txqueuelen 0  (Ethernet)
    RX packets 74  bytes 7963 (7.9 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@fed47278651b:/#

```


Lets find Mac address for Host A, Host B: **Host A's MAC is "02:42:0a:09:00:05", Host B's MAC is "02:42:0a:09:00:06"**

```
seed@VM: ~/.../Labsetup
[09/07/25]seed@VM:~/.../Labsetup$ docksh 31
root@31508b9ce735:/# arp -an
root@31508b9ce735:/# ping 10.9.0.6
PING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.
64 bytes from 10.9.0.6: icmp_seq=1 ttl=64 time=0.096 ms
64 bytes from 10.9.0.6: icmp_seq=2 ttl=64 time=0.065 ms
64 bytes from 10.9.0.6: icmp_seq=3 ttl=64 time=0.215 ms
64 bytes from 10.9.0.6: icmp_seq=4 ttl=64 time=0.101 ms
64 bytes from 10.9.0.6: icmp_seq=5 ttl=64 time=0.108 ms
^Z
[1]+  Stopped                  ping 10.9.0.6
root@31508b9ce735:/# arp -an
? (10.9.0.6) at 02:42:0a:09:00:06 [ether] on eth0
root@31508b9ce735:/# arp -n
Address                  HWtype  HWaddress                     Flags Mask
seed@VM: ~/.../Labsetup
[09/07/25]seed@VM:~/.../Labsetup$ docksh 1c
root@1c7f4b0f814c:/# arp -an
root@1c7f4b0f814c:/# ping 10.9.0.5
PING 10.9.0.5 (10.9.0.5) 56(84) bytes of data.
64 bytes from 10.9.0.5: icmp_seq=1 ttl=64 time=0.056 ms
64 bytes from 10.9.0.5: icmp_seq=2 ttl=64 time=0.057 ms
64 bytes from 10.9.0.5: icmp_seq=3 ttl=64 time=0.124 ms
64 bytes from 10.9.0.5: icmp_seq=4 ttl=64 time=0.107 ms
64 bytes from 10.9.0.5: icmp_seq=5 ttl=64 time=0.056 ms
^Z
[1]+  Stopped                  ping 10.9.0.5
root@1c7f4b0f814c:/# arp -an
? (10.9.0.5) at 02:42:0a:09:00:05 [ether] on eth0
root@1c7f4b0f814c:/#
```

Task 1.A (using ARP request): On host M, construct an ARP request packet to map B's IP address to M's MACaddress. Send the packet to A and check whether the attack is successful or not

Let's create a python file called **arp_poison.py** through Attacker machine, volumes folder

Save and execute this command in M" machine by using **chmod a+x arp_poision.py**. Now open the Host A and run the command arp -n

```
seed@VM: ~/.../Labsetup
[09/08/25]seed@VM:~/.../Labsetup$ dockps
fed47278651b  M-10.9.0.105
1c7f4b0f814c  B-10.9.0.6
31508b9ce735  A-10.9.0.5
[09/08/25]seed@VM:~/.../Labsetup$ docksh fe
root@fed47278651b:/# cd volumes/
root@fed47278651b:/volumes# nano arp_poision.py
root@fed47278651b:/volumes# chmod a+x arp_poision.py
root@fed47278651b:/volumes# arp_poision.py
.
Sent 1 packets.
root@fed47278651b:/volumes#
```



```
seed@VM: ~/.../Labsetup
File Edit View Search Terminal Help
Address          HWtype  HWaddress      Flags Mask
10.9.0.6          ether   02:42:0a:09:00:06  C
root@31508b9ce735:/# arp -n
Address          HWtype  HWaddress      Flags Mask
10.9.0.6          ether   02:42:0a:09:00:06  C
root@31508b9ce735:/# arp -n
Address          HWtype  HWaddress      Flags Mask
10.9.0.6          ether   02:42:0a:09:00:06  C
root@31508b9ce735:/# arp -n
Address          HWtype  HWaddress      Flags Mask
10.9.0.6          ether   02:42:0a:09:00:69  C
root@31508b9ce735:/#
```

From the output we can see that Host B's mac address was replaced with M's Mac.

Task 1.B (using ARP reply) :Scenario 1: B's IP is already in A's cache

Create a **arp_scenario1.py** file in attacker machine and execute that command. After the execution, Here from the output we see that the attack was **succeed**. Host A will accept the unsolicited reply and create the poisoned cache entry: 10.9.0.6 (B) -> 02:42:0a:09:00:69 (M).

```
seed@VM: ~/.../Labsetup
root@fed47278651b:/volumes# nano arp_scenario.py
root@fed47278651b:/volumes# nano arp_scenario1.py
root@fed47278651b:/volumes# chmod a+x arp_scenario1.py
root@fed47278651b:/volumes# arp-_scenario1.py
bash: arp-_scenario1.py: command not found
root@fed47278651b:/volumes# arp_scenario1.py
.
Sent 1 packets.
root@fed47278651b:/volumes# arp_scenario1.py
.
Sent 1 packets.
root@fed47278651b:/volumes# arp_scenario1.py
.
Sent 1 packets.
root@fed47278651b:/volumes#
```

```
seed@VM: ~/.../Labsetup
File Edit View Search Terminal Help
PING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.
64 bytes from 10.9.0.6: icmp_seq=1 ttl=64 time=0.230 ms
64 bytes from 10.9.0.6: icmp_seq=2 ttl=64 time=0.107 ms

--- 10.9.0.6 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1025ms
rtt min/avg/max/mdev = 0.107/0.168/0.230/0.061 ms
root@31508b9ce735:/# arp -n
Address          HWtype  HWaddress      Flags Mask
10.9.0.6          ether    02:42:0a:09:00:06  C
root@31508b9ce735:/# arp -n
Address          HWtype  HWaddress      Flags Mask
10.9.0.6          ether    02:42:0a:09:00:69  C
root@31508b9ce735:/#
```

1.B. Scenario 2: B's IP is not in A's cache.

1: Clear the cache in Host A, delete any existing entry for Host B.

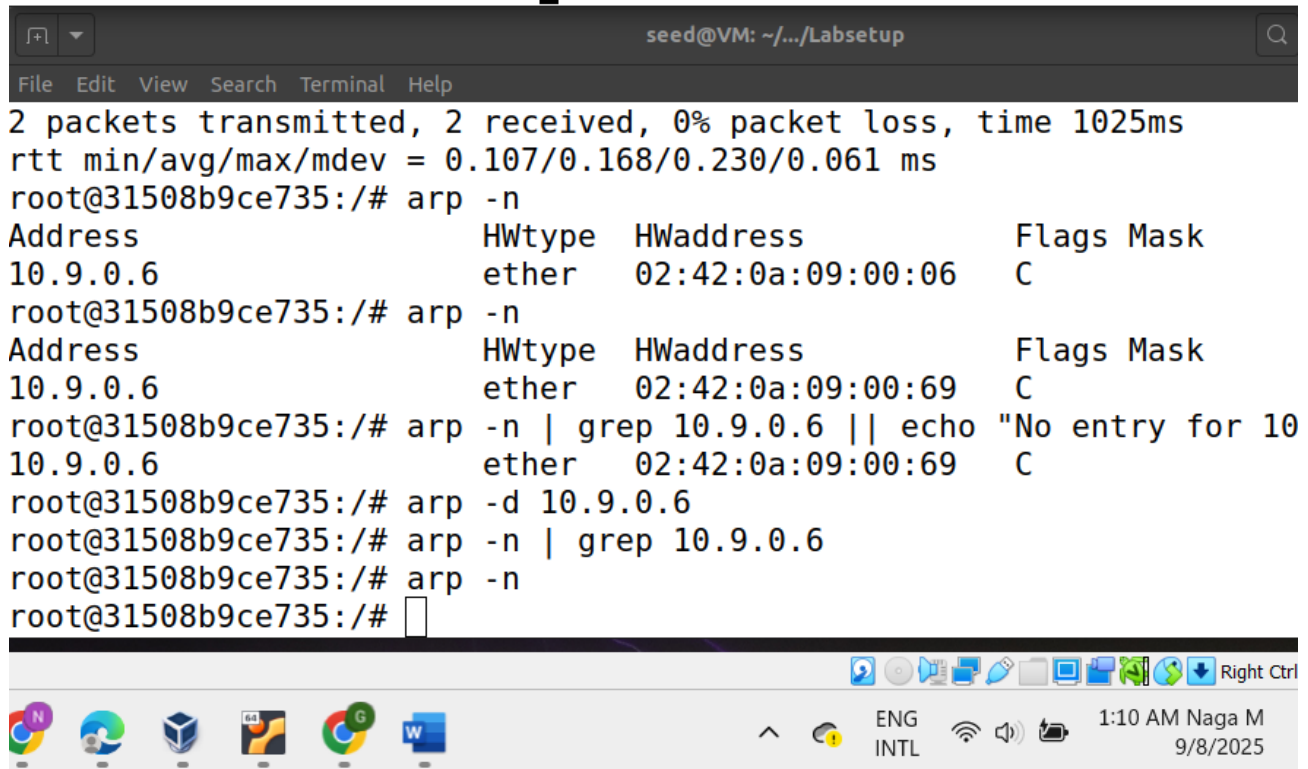
```
ipyp root@31508b9ce735:/# arp -n
Address          HWtype  HWaddress      Flags Mask
10.9.0.6          ether    02:42:0a:09:00:06  C
root@31508b9ce735:/# arp -n
Address          HWtype  HWaddress      Flags Mask
10.9.0.6          ether    02:42:0a:09:00:06  C
root@31508b9ce735:/# arp -n | grep 10.9.0.6
10.9.0.6          ether    02:42:0a:09:00:06  C
root@31508b9ce735:/# arp -d 10.9.0.6
root@31508b9ce735:/# arp -n
root@31508b9ce735:/#
```


Again run the arp_scenario1.py file in attacker machine and run the arp command in Host A, we didn't get any result.

```
root@fed47278651b:/volumes# arp_scenario1.py
```

```
.  
Sent 1 packets.
```

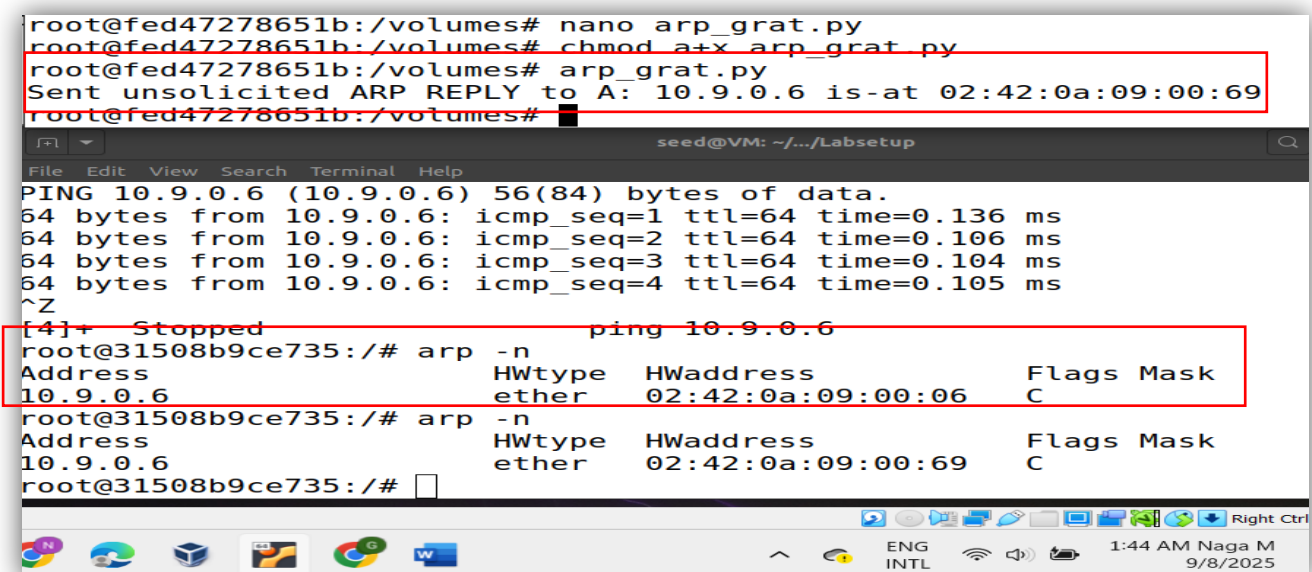
```
root@fed47278651b:/volumes#
```



```
seed@VM: ~/.../Labsetup  
File Edit View Search Terminal Help  
2 packets transmitted, 2 received, 0% packet loss, time 1025ms  
rtt min/avg/max/mdev = 0.107/0.168/0.230/0.061 ms  
root@31508b9ce735:/# arp -n  
Address HWtype HWaddress Flags Mask  
10.9.0.6 ether 02:42:0a:09:00:06 C  
root@31508b9ce735:/# arp -n  
Address HWtype HWaddress Flags Mask  
10.9.0.6 ether 02:42:0a:09:00:69 C  
root@31508b9ce735:/# arp -n | grep 10.9.0.6 || echo "No entry for 10  
10.9.0.6 ether 02:42:0a:09:00:69 C  
root@31508b9ce735:/# arp -d 10.9.0.6  
root@31508b9ce735:/# arp -n | grep 10.9.0.6  
root@31508b9ce735:/# arp -n  
root@31508b9ce735:/#
```

The screenshot shows a terminal window with a Windows taskbar at the bottom. The taskbar includes icons for various applications and system status indicators like language (ENG/INTL), network, and battery. The system clock shows 1:10 AM on 9/8/2025.

Task 1.C (using ARP gratuitous message): create a arp_grat.py file and execute that and run the arp command in Host A

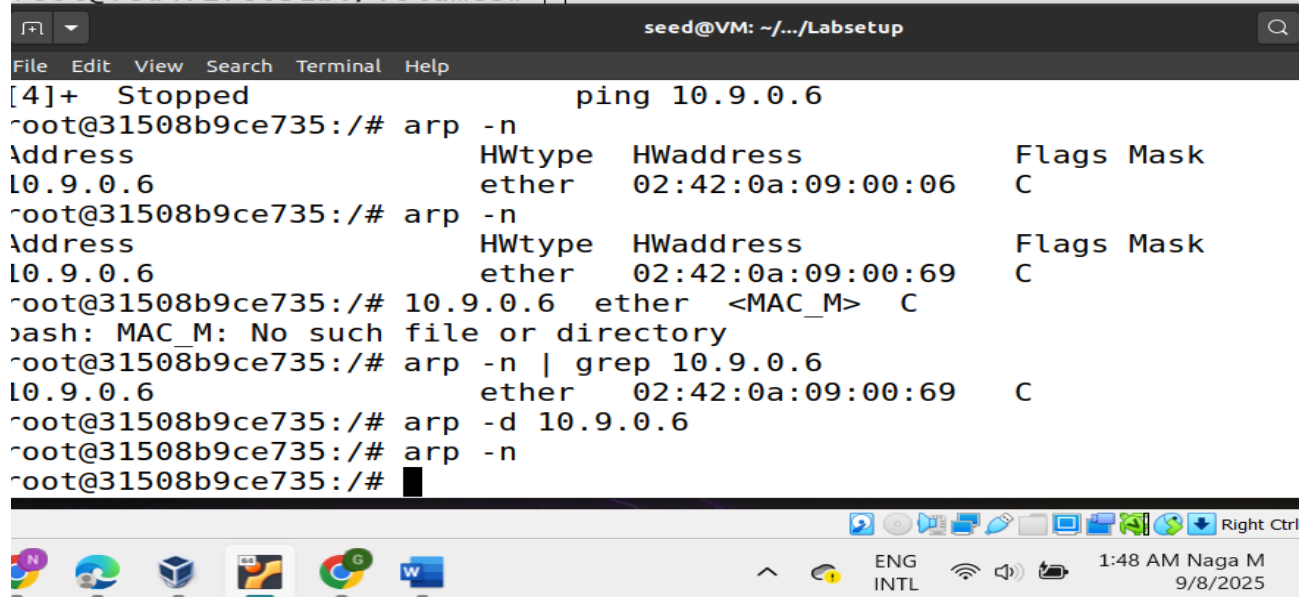


```
root@fed47278651b:/volumes# nano arp_grat.py  
root@fed47278651b:/volumes# chmod a+x arp_grat.py  
root@fed47278651b:/volumes# arp_grat.py  
Sent unsolicited ARP REPLY to A: 10.9.0.6 is-at 02:42:0a:09:00:69  
root@fed47278651b:/volumes#  
seed@VM: ~/.../Labsetup  
File Edit View Search Terminal Help  
PING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.  
64 bytes from 10.9.0.6: icmp_seq=1 ttl=64 time=0.136 ms  
64 bytes from 10.9.0.6: icmp_seq=2 ttl=64 time=0.106 ms  
64 bytes from 10.9.0.6: icmp_seq=3 ttl=64 time=0.104 ms  
64 bytes from 10.9.0.6: icmp_seq=4 ttl=64 time=0.105 ms  
^Z  
[4]+ Stopped ping 10.9.0.6  
root@31508b9ce735:/# arp -n  
Address HWtype HWaddress Flags Mask  
10.9.0.6 ether 02:42:0a:09:00:06 C  
root@31508b9ce735:/# arp -n  
Address HWtype HWaddress Flags Mask  
10.9.0.6 ether 02:42:0a:09:00:69 C  
root@31508b9ce735:/#
```

The screenshot shows a terminal window with a Windows taskbar at the bottom. The taskbar includes icons for various applications and system status indicators like language (ENG/INTL), network, and battery. The system clock shows 1:44 AM on 9/8/2025.

Scenario2 :

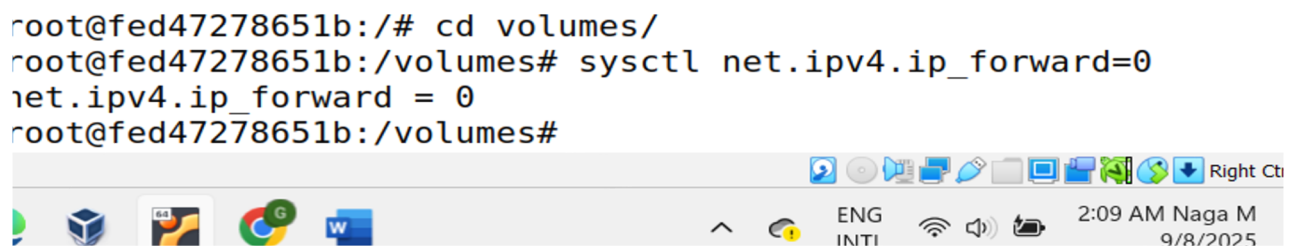
```
root@fed47278651b:/volumes# arp_grat.py
Sent unsolicited ARP REPLY to A: 10.9.0.6 is-at 02:42:0a:09:00:69
root@fed47278651b:/volumes#
```



```
File Edit View Search Terminal Help
[4]+ Stopped ping 10.9.0.6
root@31508b9ce735:/# arp -n
Address HWtype HWaddress Flags Mask
10.9.0.6 ether 02:42:0a:09:00:06 C
root@31508b9ce735:/# arp -n
Address HWtype HWaddress Flags Mask
10.9.0.6 ether 02:42:0a:09:00:69 C
root@31508b9ce735:/# 10.9.0.6 ether <MAC_M> C
bash: MAC_M: No such file or directory
root@31508b9ce735:/# arp -n | grep 10.9.0.6
10.9.0.6 ether 02:42:0a:09:00:69 C
root@31508b9ce735:/# arp -d 10.9.0.6
root@31508b9ce735:/# arp -n
root@31508b9ce735:/#
```

TASK 2:

We need to create `arp_mitm.py` file and run that, open another terminal in `M'`. and in that root execute the following command:



```
root@fed47278651b:/# cd volumes/
root@fed47278651b:/volumes# sysctl net.ipv4.ip_forward=0
net.ipv4.ip_forward = 0
root@fed47278651b:/volumes#
```

Ping Host B from Host A, and see the output from the below, It should **fail** (packets now arrive at M and stop).

```
seed@VM: ~/.../Labsetup
seed@VM: ~/.../Labsetup
root@fed47278651b:/volumes# arp_mitm.py
^Z
[1]+  Stopped                  arp_mitm.py
root@fed47278651b:/volumes# arp_mitm.py
^Z
[2]+  Stopped                  arp_mitm.py
root@fed47278651b:/volumes# chmod a+x arp_mitm.py
root@fed47278651b:/volumes# arp_mitm.py
^Z
[3]+  Stopped                  arp_mitm.py
root@fed47278651b:/volumes# nano arp_mitm.py
root@fed47278651b:/volumes# chmod a+x arp_mitm.py
root@fed47278651b:/volumes# arp_mitm.py
[6]+  Stopped                  ping 10.9.0.6
root@31508b9ce735:/# ping 10.9.0.6
PING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.
64 bytes from 10.9.0.6: icmp_seq=1 ttl=64 time=0.160 ms
64 bytes from 10.9.0.6: icmp_seq=2 ttl=64 time=0.110 ms
64 bytes from 10.9.0.6: icmp_seq=3 ttl=64 time=0.105 ms
64 bytes from 10.9.0.6: icmp_seq=4 ttl=64 time=0.122 ms
64 bytes from 10.9.0.6: icmp_seq=5 ttl=64 time=0.088 ms
64 bytes from 10.9.0.6: icmp_seq=6 ttl=64 time=0.108 ms
^Z
[7]+  Stopped                  ping 10.9.0.6
root@31508b9ce735:/# ping 10.9.0.6
PING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.
```

Lets change the sysctl net. ipv4.ip_forward=1, run the same above. It will forward that packets.

Step4:

```
root@fed47278651b:/volumes# nano arp_telnet.py
ar
root@fed47278651b:/volumes# chmod arp_telnet
chmod: missing operand after 'arp_telnet'
Try 'chmod --help' for more information.
root@fed47278651b:/volumes# chmod a+x arp_telnet.py
root@fed47278651b:/volumes# arp_telnet.py
[*] Using iface: eth0
[*] BPF filter:
    tcp and (ether src 02:42:0a:09:00:05 or ether src 02:42:0a:09:00:06 ) or tcp and
rc host 10.9.0.5 or dst host 10.9.0.6 )
[*] TIP: keep your ARP-poisoning running separately, and set ip_forward=0 on M.
```

In Host A, type telnet 10.9.0.6

You will get in to Host B screen appeared in Host A because of telnet communication:



```
seed@VM: ~/.../Labsetup
telnet> exit
Invalid command
telnet> ^Z
[10]+  Stopped                  telnet 10.9.0.6
root@31508b9ce735:/# telnet 10.9.0.6
Trying 10.9.0.6...
Connected to 10.9.0.6.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
1c7f4b0f814c login: seed
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-54-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

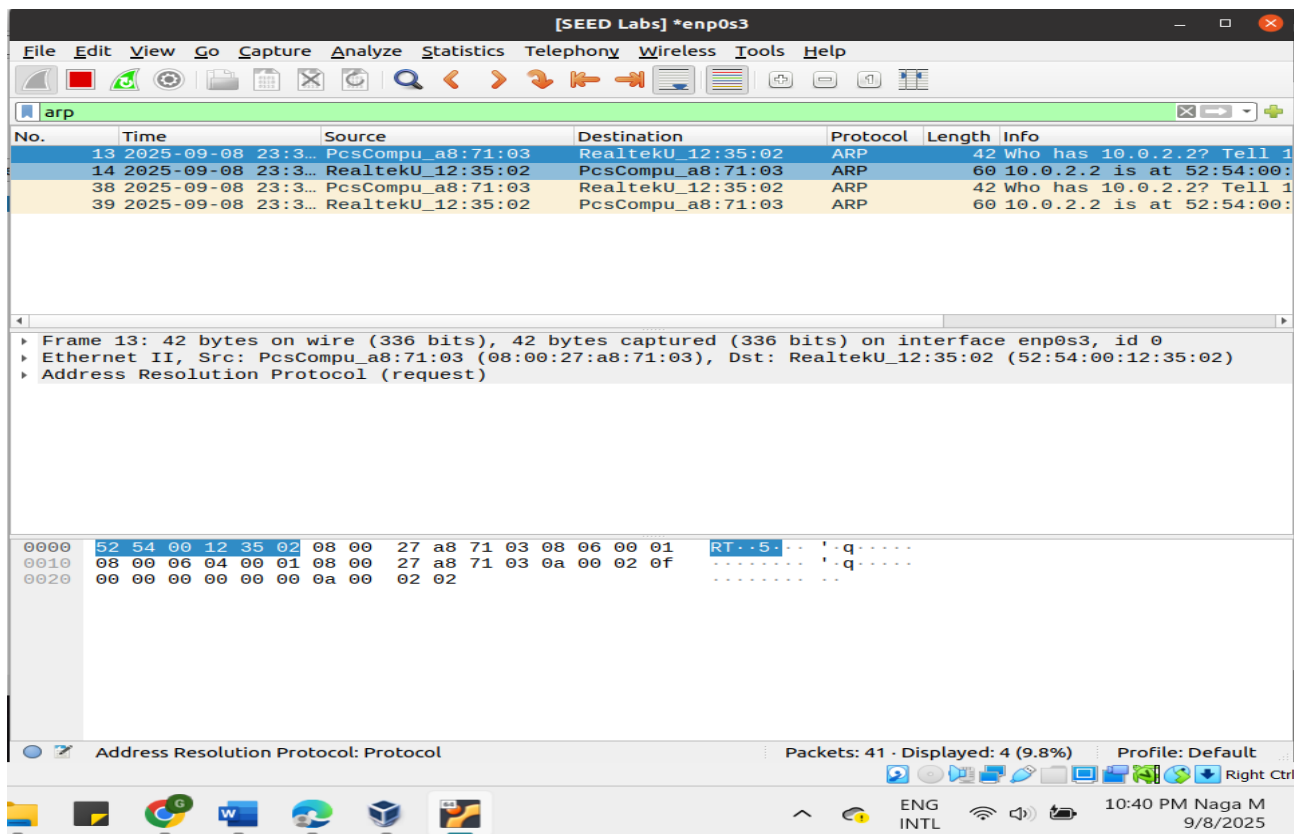
To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

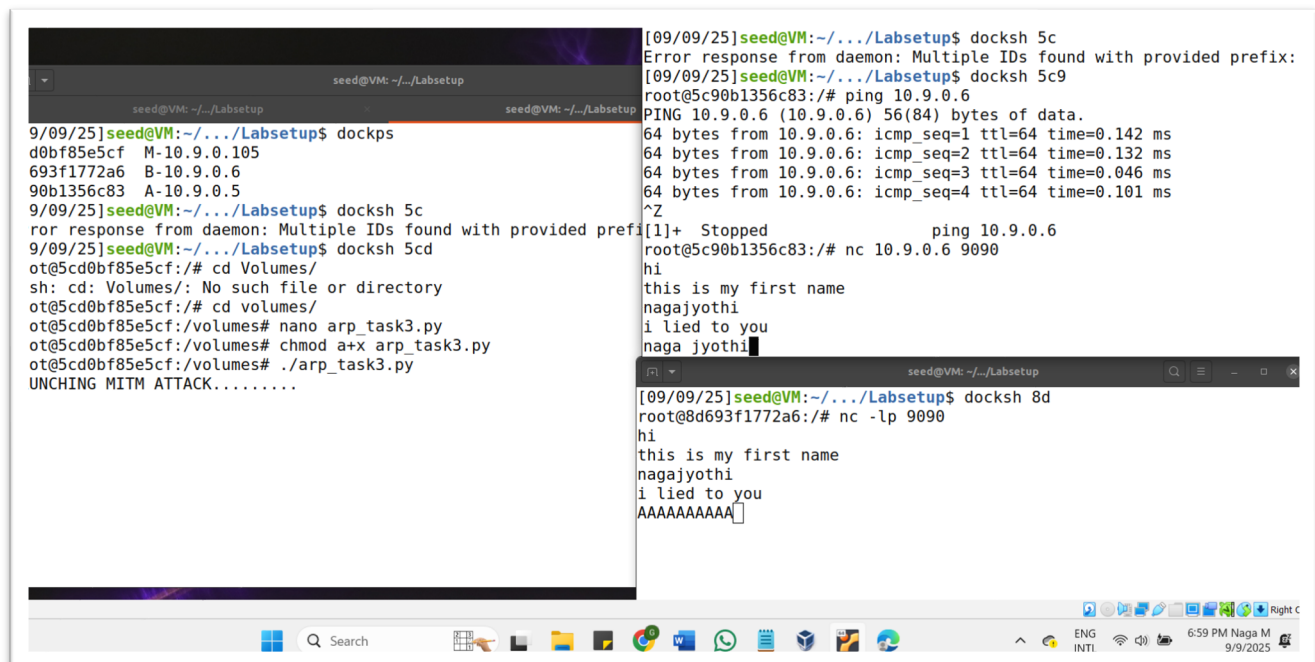
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

seed@1c7f4b0f814c:~$
[4]+  Stopped                  arp_mitm.py
root@fed47278651b:/volumes# nano arp_mitm.py
root@fed47278651b:/volumes# arp_mitm.py
[8]+  Stopped                  ping 10.9.0.6
root@31508b9ce735:/# ping 10.9.0.6
PING 10.9.0.6 (10.9.0.6) 56(84) bytes of data.
64 bytes from 10.9.0.6: icmp_seq=1 ttl=63 time=0.130 ms
64 bytes from 10.9.0.6: icmp_seq=2 ttl=63 time=0.152 ms
From 10.9.0.105: icmp_seq=3 Redirect Host(New nexthop: 10.9.0.6)
64 bytes from 10.9.0.6: icmp_seq=3 ttl=63 time=0.104 ms
From 10.9.0.105: icmp_seq=4 Redirect Host(New nexthop: 10.9.0.6)
64 bytes from 10.9.0.6: icmp_seq=4 ttl=63 time=0.202 ms
```

From the wire shark, we can see that ARP packets are sending



Task3:



Lets execute the arp_task3.py in attacker machine. Here, from this output we can see that there , the attacker machine interrupted the communication between Host A and Host B, and replace the first name with all AAAAAA's.